APOPHIS 2029: PREDICTION RECORD

Project Hannibal study file · record of two competing predictions for the 13 April 2029 close approach of asteroid 99942 Apophis · prepared 23 October 2026

This document records two predictions, made before the event, for how close Apophis will come to Earth on 13 April 2029. The authoritative record is the git history of the repository below; this PDF is a readable copy of it. Both predictions are timestamped by commit and are scored against the reconstructed orbit published after the flyby.

1 The predictions

Party	Prediction (distance from Earth's centre at closest approach)	Basis
Bram, investigator, Project Hannibal	**28,000 to 29,000 km** ("the model is off")	The investigators' hypothesis
Claude, AI assistant (Anthropic)	**38,008 to 38,015 km**, closest approach at 21:45 UTC ± 1 minute	NASA/JPL orbit solution 220, 3-sigma window, confirmed independently by ESA NEOCC

Investigator's stated reason (made public at his request, commit 69628c9): 28,000 to 29,000 km is the orbital shell of the medium-orbit navigation satellites (BeiDou about 27,900 km and Galileo about 29,600 km from Earth's centre; GPS and GLONASS just below). The prediction is that Apophis passes through that shell rather than between it and the geostationary ring, where both agencies place it. He describes it as a bet he hopes to lose.

Recorded in `docs/investigation/apophis/PREREGISTRATION.md`, section 4d, commit `d0b7730` (23 Oct 2026, 21:46:07 UTC). Stake: reputation only. No money, goods or services are involved.

2 Scoring rules (fixed in advance)

Outcome (reconstructed closest-approach distance)	Result
38,008 to 38,015 km	Claude's prediction correct; the orbit models confirmed
28,000 to 29,000 km	Bram's prediction correct; the orbit models were wrong
Anywhere else	Neither wins. A value outside 38,008 to 38,015 km still shows the models were wrong.

Deciding source: NASA/JPL's post-flyby orbit solution (JPL Small-Body Database / CNEOS close-approach data), checked against ESA NEOCC's reconstruction. If the two disagree by more than their stated uncertainties, both values are reported and the record says so.

Both predictions are misses. At 28,000 km Apophis would still pass about 21,500 km above Earth's surface. Neither prediction describes an impact.

3 Data the predictions were made against

Quantity (as of 23 Oct 2026)	NASA/JPL	ESA NEOCC
Orbit solution	220 (condition code 0), 7,370 observations, last 2022-04-09	independent fit, 9,577 observations on file, last 2022-04-09
2029 closest approach	38,011 km; 3-sigma window 38,008 to 38,015 km; 21:44:50 UTC	38,014 km (1-sigma 1.3 km); 21:45:03 UTC
Agreement	2.3 km apart = 1.4 combined sigma	
Yarkovsky push A2	−2.9018e−14 ± 1.9e−16 au/d²	−2.9001e−14 ± 2.3e−16 au/d²
Impact-risk list	Sentry: removed 2021-02-21	not listed

Raw replies from both agencies, fetched live and hashed, are in `docs/investigation/apophis/evidence/2026-10-23_2103/`:

```
fc23f64a…bfb024 jpl_cad.json
ee5ae262…13b373fc jpl_sentry.json
4b7720e8…6f552f jpl_sbdb.json
bce0af62…cee5ebe8 esa_clolin.txt
a51f86d0…47e7465d manifest.json
```

4 What would show up before 2029

Moving the closest approach by 9,000 to 10,000 km is several thousand times JPL's stated uncertainty, so it would not stay hidden until the day. The study's daily brief (three times a day) raises an alert on the first new orbit solution that moves the 2029 distance by more than 3.3 km (criterion C5), or if a push the Yarkovsky model doesn't explain is measured (C6). Apophis is observable again from about 20 December 2026, and every night from 7 October 2028 to the flyby.

5 Audit trail

Commit	Time (UTC)	What it recorded
36f3703	2026-10-23 21:03 UTC	Pre-registration, baseline, JPL vs ESA cross-check, evidence run 2026-10-23_2103
5ed4897	2026-10-23 21:24 UTC	Hypothesis H-A and criteria C6 to C8
5466d4a	2026-10-23 21:30 UTC	Worst-case rule
65431af	2026-10-23 21:42 UTC	2029 model test: JPL and ESA predictions frozen
d0b7730	2026-10-23 21:46 UTC	The bet recorded (section 4d)
8bf1987	2026-10-23 21:47 UTC	Section headings cite their commits
e243abb	2026-10-23 22:34 UTC	Criterion C9: photometric distance check
69628c9	2026-10-23 22:53 UTC	Investigator's reason for his range made public (4d addendum)

Repository: github.com/3mm4nu3l/Hannibal-Beta-2.0, branch reports/daily (pull request #15). To verify: `git show d0b7730:docs/investigation/apophis/PREREGISTRATION.md`; evidence: `cd docs/investigation/apophis/evidence/2026-10-23_2103 && sha256sum -c SHA256SUMS`.

6 Disclosures and compliance notes

Item	Statement
Nature	A scientific prediction record between an investigator and an AI assistant. Not a wager: no money or other consideration changes hands.
AI involvement	Claude (Anthropic) built the analysis software and wrote this record at the investigator's request. Its prediction comes from public agency data, not independent knowledge.
Data sources	Public NASA/JPL APIs (ssd-api.jpl.nasa.gov, ssd.jpl.nasa.gov/api/horizons.api) and ESA NEOCC public downloads (neo.ssa.esa.int). No authentication, no restricted data.
Personal data	None beyond the investigator's first name. No location, contact or identity data of third parties.
Activities	Receive-only. No radio transmission. Any optical or radio observations are passive.
Limitations	Agency figures use linearised 3-sigma uncertainties. The scoring depends on the post-flyby reconstruction, which will itself carry an uncertainty of order kilometres.
Pre-registration	Criteria and predictions were committed before any data testing them existed. Later changes are new commits with a stated reason.

Prepared from the repository state at commit 69628c9 (2026-10-23 22:53 UTC). If this PDF and the repository ever disagree, the repository's git history is authoritative.